

Sofia Nikiforova

Applied AI/ML Engineer & Data Scientist

NLP · LLMs · 8+ Years Python

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Applied AI/ML engineer and data scientist specializing in NLP, LLMs, and Generative AI, with 8+ years of professional Python experience and a Ph.D. in Computational Linguistics/NLP.

I design, build, evaluate, and improve AI systems for real-world use, spanning LLM-powered data analysis, information extraction, text generation, multimodal AI, and predictive modeling. My work emphasizes model behavior, reliable evaluation workflows, error analysis, and measurable system improvement.

Experience

Senior AI Researcher

Oct 2024–June 2026

Causify

Led applied AI projects developing LLM-powered data analysis tools, causal inference systems, and time-series prediction solutions, with ownership across modeling, evaluation, stakeholder communication, and production-oriented delivery.

- Led development of an LLM-powered Python tool for generating, validating, and interacting with causal DAGs from datasets and metadata. Used the OpenAI API, structured outputs, and Pydantic-based validation to support reliable graph generation, user feedback, and on-demand causal analysis.
- Led development of a causal failure prediction framework for wind turbines, combining sensor time series data, domain knowledge, and predictive modeling to identify early indicators of equipment failure.
- Supervised and evaluated interns working on AI, data science, and causal inference projects.
- Contributed to research papers, technical reports, patent-related work, and internal engineering practices, including pytest-based testing, Git/GitHub workflows, and automated CI checks with GitHub Actions.

Postdoctoral Researcher

Mar 2023–Oct 2024

Utrecht University

Developed multimodal image-text AI systems for radiology, combining CT imaging data with semi-structured radiology report texts for clinical decision support.

- Used locally hosted Llama models via Ollama for NLP-based information extraction from radiology reports, deriving structured clinical context and aligning it with visual findings from medical images.
- Evaluated outputs for factual accuracy, interpretability, and robustness in a safety-sensitive healthcare setting.
- Collaborated with an interdisciplinary team across AI, medicine, and engineering.

Senior Data Scientist

Nov 2016–Jun 2021

Particle.One

Led development of Python-based NLP pipelines for text classification, information extraction, and predictive analytics in the financial domain.

- Owned applied NLP workflows from problem formulation and data acquisition through feature engineering, model training, evaluation, deployment support, monitoring, and iterative improvement.
- Collaborated with business and engineering stakeholders to translate data problems into production-oriented ML solutions.

Teaching Assistant

Sep 2021–Nov 2021

Utrecht University

Course: Machine Learning for Human Vision and Language (MSc Artificial Intelligence).

Developed and graded assignments and supervised practical lab sessions covering state-of-the-art machine learning techniques for vision and language tasks.

NLP Developer

Jan 2017–Jan 2018

Semantic Hub

Developed NLP tools for information extraction in the life sciences/biomedical domain. Implemented pipelines for entity recognition, relation extraction, and structured data generation from heterogeneous unstructured text, including technical documents and user-generated content.

Education

Ph.D., Computational Linguistics/NLP

2018–Feb 2023

Utrecht University, Netherlands

Dissertation: *Image Captioning with External Knowledge*

- Developed transformer-based knowledge-grounded image captioning models in PyTorch.

- Designed and implemented models that identified relevant entities, retrieved contextual information, and incorporated that information into generated image descriptions.
- Focused on factual accuracy, grounding, dataset creation, and systematic evaluation across geographic and news-image domains.

M.A., Computational Linguistics

Higher School of Economics, Moscow, Russia

2016–2018

B.A., Fundamental and Applied Linguistics

Lomonosov Moscow State University (MSU), Russia

2012–2016

Selected Projects

Contract RAG Agent [GitHub repository](#)

Built an agentic retrieval-augmented generation system for question answering over contract texts, with safety checks, document/chunk retrieval, grounded answer generation with supporting quotes, answer verification, and failure handling for unsupported answers or insufficient evidence. Implemented in Python with LangGraph/LangChain, FastAPI, Elasticsearch, Pydantic, SQLite, and Docker Compose.

Selected Publications & Patents

- **DMCD: Semantic-Statistical Framework for Causal Discovery.** *arXiv preprint, 2026.* [arXiv:2602.20333](#)
A two-phase causal discovery framework that integrates LLM-based semantic drafting from variable metadata with statistical validation on observational data.
- **Generating Image Captions with External Encyclopedic Knowledge.** *arXiv preprint, 2022.* [arXiv:2210.04806](#)
Image captioning approach that integrates encyclopedic knowledge into generation, with emphasis on factual accuracy.
- **Geo-Aware Image Caption Generation.** *Proceedings of COLING 2020.* [ACL Anthology](#)
Image captioning with geographic context grounding for generating location-aware photo descriptions.
- **Systems and Methods for Causal Failure Prediction in Time Series** U.S. Patent Application No. 63/934,790

Skills & Technologies

Skills

Natural Language Processing (NLP) · Large Language Models (LLMs) · Generative AI · LLM Evaluation · Model Evaluation · Error Analysis · Retrieval-Augmented Generation (RAG) · Prompt Engineering · Structured Outputs · Agentic Workflows · Information Extraction · Information Retrieval · Dataset Creation · Experiment Design · Explainability · Multimodal AI · Vision-Language Models · Conversational AI · Machine Learning · Causal Inference · Time-Series Modeling

Professional Strengths

Analytical Problem-Solving · Independent Project Ownership · Stakeholder Communication · Cross-Functional Collaboration · Technical Leadership · Agile/Scrum Collaboration

Technologies

Python · OpenAI API · Ollama · LangGraph · LangChain · PyTorch · scikit-learn · pandas · NumPy · Elasticsearch · Pydantic · pytest · Git · GitHub · GitHub Actions · CI Workflows · Docker · FastAPI · SQLite · Jupyter · Linux · LaTeX

Languages

English (fluent) · Dutch (upper-intermediate) · Russian (native)

Location & Availability

Amersfoort, Netherlands · **Dutch citizen** · No sponsorship required · **Available immediately**